

Machine, Data and Learning

Utility Theory + Decision Theory
Chapter 17 from the book by Russell
and Norvig

Decision Theory

(How to make decisions)

Decision Theory

= Probability theory + Utility Theory
(deals with chance) *(deals with outcomes)*

- *Fundamental idea:*
 - The **MEU** (Maximum expected utility) principle
 - Agent is **rational** if and only if it chooses the action that yields the highest expected utility, averaged over all possible outcomes of the action
 - Weigh the utility of each outcome by the probability that it occurs

Revisiting Romania example

- If plan1 and plan2 are the two plans:
 - Plan 1 uses route 1
 - $P(\text{home-early} | \text{plan1}) = .8$, while $P(\text{stuck1} | \text{plan1}) = .2$
 - Route 1 will be quick if flowing, but stuck for 1 hour if slow
 - $U(\text{home-early}) = 100$, $U(\text{stuck1}) = -1000$
 - Assigned numerical values to outcomes!
 - Plan 2 uses route 2
 - $P(\text{home-somewhat-early} | \text{plan2}) = .7$, $P(\text{stuck2} | \text{plan2}) = .3$
 - Route 2 will be somewhat quick if flowing, but not bad even if slow
 - $U(\text{home-somewhat-early}) = 50$, $U(\text{stuck2}) = -10$

Application of MEU Principle

- $$\begin{aligned} EU(\text{Plan1}) &= P(\text{home-early} \mid \text{plan1}) * U(\text{home-early}) \\ &\quad + P(\text{stuck1} \mid \text{plan1}) * U(\text{stuck1}) \\ &= 0.8 * 100 + 0.2 * -1000 = -120 \end{aligned}$$
- $$\begin{aligned} EU(\text{Plan2}) &= P(\text{home-somewhat-early} \mid \text{plan2}) * U(\text{home-somewhat-early}) \\ &\quad + P(\text{stuck2} \mid \text{plan2}) * U(\text{stuck2}) \\ &= 0.7 * 50 + 0.3 * -10 = 32 \end{aligned}$$

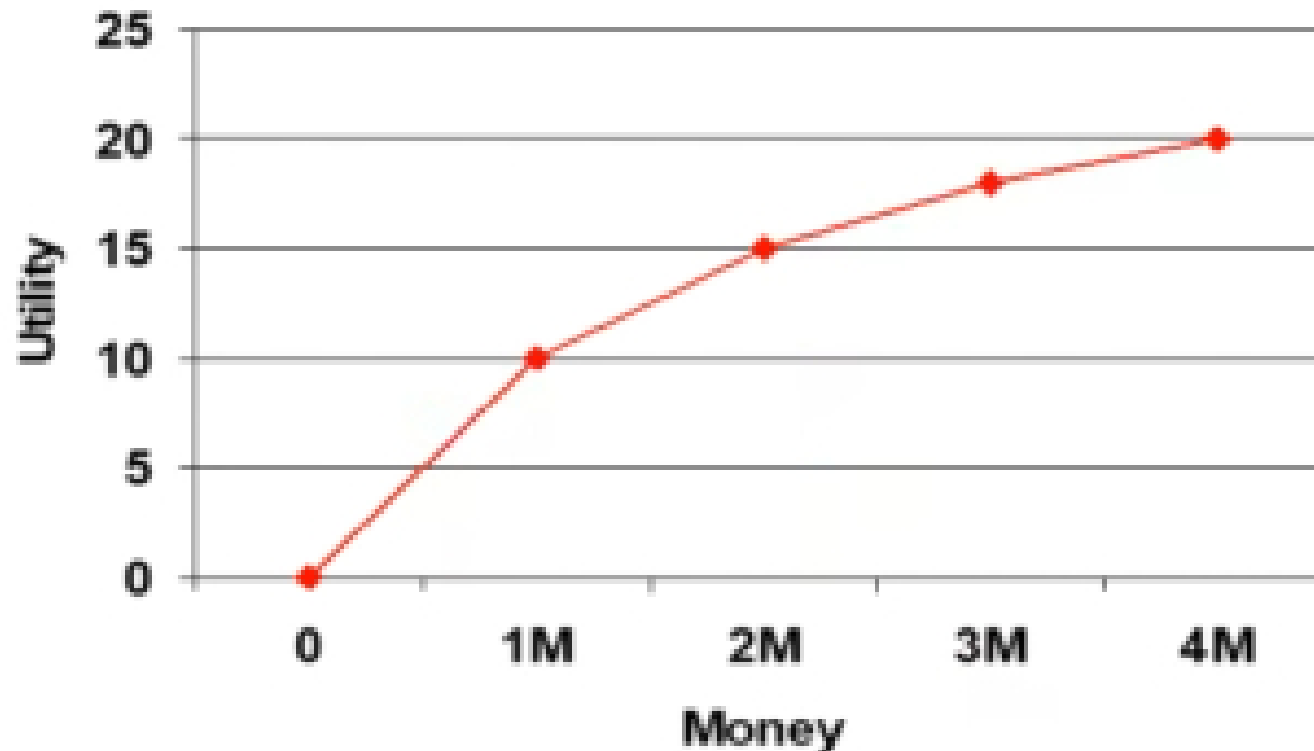
Lottery Example

- Suppose an agent gives you a choice:
 - **Choice 1:** You will get \$1,000,000
 - **Choice 2:** The agent will toss a coin
 - If heads, then you win \$3,000,000
 - If tails, then you get nothing
- Simple expected utility calculations give:
 - $EU(\text{Choice1}) = \$1,000,000$
 - $EU(\text{Choice2}) = \$1,500,000$
- So why did we prefer the first choice?

Risk Aversion

- We are **risk averse**
- Our utility functions for money are as follows (!!):
 - Our first million means a lot $U(\$1M) = 10$
 - Second million not so much $U(\$2M) = 15$ (NOT 20)
 - Third million even less so $U(\$3M) = 18$ (NOT 30)
 -
- Additional money is not buying us as much utility
- If we plot amount of money on the x-axis and utility on the y-axis, we get a concave curve

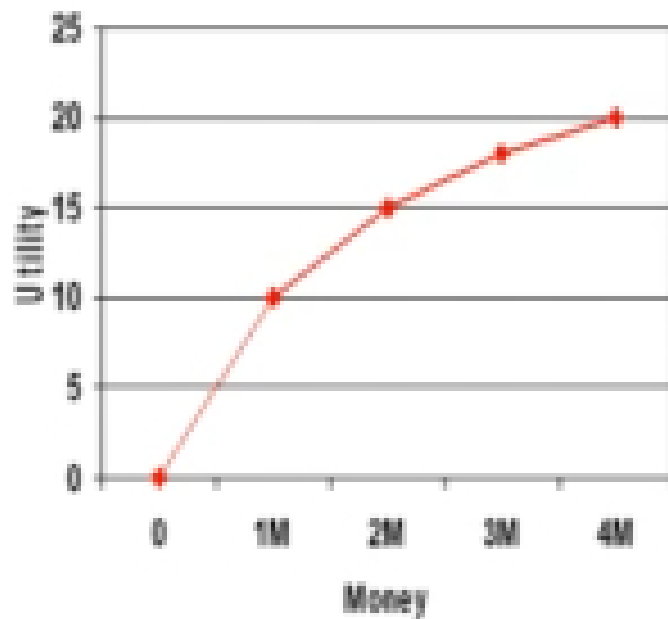
Answer: Risk Aversion



- $EU(\text{choice1}) = U(\$1\text{M}) = 10$
- $EU(\text{choice2}) = 0.5 * U(0) + 0.5 * U(\$3\text{M} = 18) = 9$
- That is why we prefer the sure \$1M

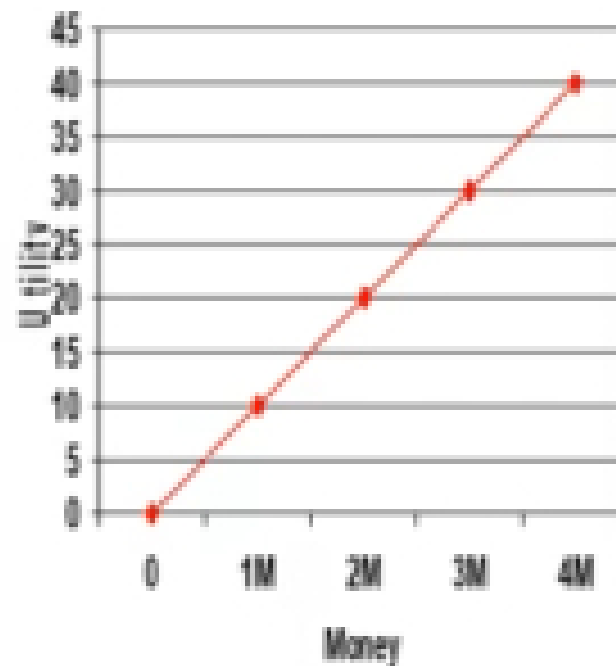
Risk Averse, Risk Neutral Risk Seeking

RISK AVERSE



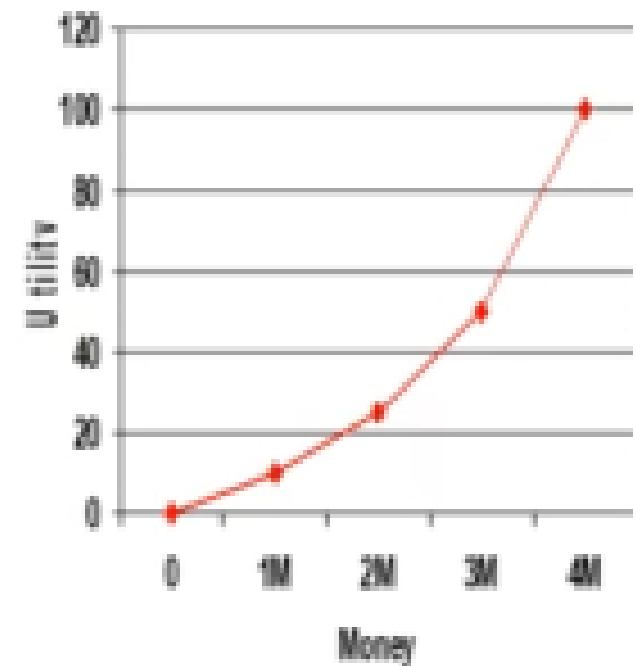
$EU(\text{Choice1}) = 10$
 $EU(\text{Choice2}) = 9$

RISK NEUTRAL



$EU(\text{Choice1}) = 10$
 $EU(\text{Choice2}) = 15$

RISK SEEKER



$EU(\text{Choice1}) = 10$
 $EU(\text{Choice2}) = 25$

More Risk Aversion

- Key: Slope of utility function is **continuously decreasing**
 - We will refuse to play a monetarily fair bet
- Suppose we start with x dollars
 - We are offered a game:
 - 0.5 chance to win 1000 dollars ($c = 1000$)
 - 0.5 chance to lose 1000 dollars ($c = 1000$)
 - Expected monetary gain or loss is zero (hence monetarily fair)
 - Should be neutral to it, but seems we are not! Why?
 - $U(x + c) - U(x) < U(x) - U(x - c)$
 - $U(x + c) + U(x - c) < 2 U(x)$
 - $[U(x + c) + U(x - c) / 2] < U(x)$
 - $EU(\text{playing the game}) < EU(\text{not playing the game})$

Multiattribute Utility Theory

- Can we capture desirability of outcomes in a single utility function?
- Suppose renting an apartment
 - House1: closer-to-university, newer, costs 100 units
 - House2: Farther-from-university, older, costs 85 units
 - (Assume, you can afford up to 100 units)
- Outcomes characterized by two or more attributes
 - Attributes: $X_1, X_2, \dots, X_N$, e.g., <distance-to-univ, old/new, cost>
 - Values: $x_1, x_2, \dots, x_N$,
 - Closer-to-univ = 1, farther-from-univ = 0; new = 1, old = 0
 - Apartment1: <1,1,-100> Apartment2: <0,0, -85>
 - Which is a better apartment? (Pairwise comparison fails)

Multiattribute Utility Theory

- Don't get a single number, but vector of values as outcomes, $\langle x, y \rangle$
- How do you compare values now?
 - Compare $\langle 1, 1, -100 \rangle$ with $\langle 0, 0, -85 \rangle$
 - Compare $\langle 3, 3, 5 \rangle$ with $\langle 5, 3, 3 \rangle$
- One approach is dominance (strict, stochastic...):
 - If you are lucky, find $\langle 3, 3, 3 \rangle$ and $\langle 3, 3, 5 \rangle$
 - Values in one vector dominate values in the other vector

Markov Decision Process (MDP)

Chapter 17: Making Complex Decisions

- Defined as a tuple: $\langle S, A, P, R \rangle$
 - **S**: State
 - **A**: Action
 - **P**: Transition function
 - Table $P(s' | s, a)$, prob of s' given action “a” in state “s”
 - **R**: Reward
 - $R(s, a)$ = cost or reward of taking action a in state s

Example: What SEQUENCE of actions should our agent take?

- Agent can take action N, E, S, W
- Each action costs $-1/25$

